

Program at a glance

Time	*Adelaide Oval*			*Flinders Festival Plaza*	
	Monday 24th	**Tuesday 25th**	**Wednesday 26th**	**Thursday 27th**	**Friday 28th**
Day	COMBINE, SBEI & the ABACBS EMCR Shark Tank	ABACBS 2025 Conference	ABACBS 2025 Conference	BioC Asia and ABACBS Workshops	BioC Asia and ABACBS Workshops
Evening	ABACBS 2025 Welcome Reception	Conference Dinner	BioC Asia & R Ladies Networking		

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ALIGN Australian Alliance for Indigenous Genomics

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AMD

MGI

Cancer Bioinformatics Australia cancerbioinformatics.au

CSL

SINCE 1874 Bickford's AUSTRALIA

Red Bull ENERGY DRINK

LONGVIEW MACCLESFIELD

Venue

ABCABS 2025 will be held at the Adelaide Oval Conference Centre (Mon–Wed). Enter via the South Gate (marked below with a blue X). This is where registration and signage will be. From the CBD, the simplest route is through the train station and across the Riverbank pedestrian bridge (~600 m, and quite a scenic walk). The ABACBS 2025 workshops and BioC Asia (Thurs-Fri) will be held at Flinders University Festival Plaza (labelled below with "workshops")



Time	Tuesday 25th (Adelaide Oval)	
09:00	**Welcome to Country with Cliffy Wilson followed by opening remarks from Her Excellency the Honourable Frances Adamson AC, Governor of South Australia**	
09:20	**EMBL-EBI Data Resources in 2025 - opportunities and challenges - Dr Joanna McEntyre. Interim Director, EMBL-EBI.** Sponsored by Adelaide University. Chair - Jimmy Breen	
10:00	**Lightning talks.** Chair - Jimmy Breen	
10:30	Morning Tea & Poster Session & Exhibitor time	
11:20	**Panel Discussion: The Landscape of Research Funding in Australia** **A discussion with leaders from NHMRC, ARC, and philanthropic funding bodies.** Chair - Jimmy Breen	
12:10	Lunch & Exhibitor Time	
13:10	***William Magarey Room - Cancer and Molecular Medicine.*** *Sponsored by Cancer Bioinformatics Australia* *Chairs - Nadia Davidson & Stefano Mangiola*	***John Halbert Room - Microbial bioinformatics.*** *Sponsored by FAME* *Chairs - Vijini Mallawaarachchi & Calum Walsh*
	Tumour subclones and their microenvironments - Prof. Alistair Forrest, Harry Perkins Institute of Medical Research, Perth.	*Implementing pathogen genomics into public health - Dr. Jessica Webb, University of Adelaide*
13:30	RobSL: A Cross-Platform Computational Framework for Robust Synthetic Lethality Discovery in Cancer - Zikang Xu	MGTdb: a web service and database for studying the global and local genomic epidemiology of bacterial pathogens - Sandeep Kaur
13:40	Towards Clinically Feasible Multi-Omics Diagnostics: A Deep Learning Framework for Low-Throughput Biomarker Discovery from Blood Biopsies - Ricky Nguyen	Barbet: A Deep Learning Approach to Bacterial Taxonomic Classification - Wytamma Wirth
13:50	Integrative Multi-Omics Framework for Causal Gene Discovery in Long COVID - Sindy Pinero	Structural and Functional Annotation of a Novel Phage Plasmid Hybrid in Bacillus anthracis - Lindsay Durr
14:00	Multi-modal analysis reveals new links between the tumour microenvironment and prostate cancer heterogeneity - William John Hutchison	Computational fitness landscape and evolutionary models for combating Hepatitis C virus (HCV) - Ahmed Abdul Quadeer
14:10	Predictive Proteomic Biomarkers Enhance Patient Stratification in Metastatic Colorectal Cancer A Post Hoc Analysis of the FIRE-3/AIO KRK-0306 Trial - Alina Jana Arneth	DAMPA - accelerated and simplified design of probe panels for targeted metagenomics using pangenome graphs - Michael Payne
14:20	Large-scale Multi-cohort Single-cell Modelling Reveals Molecular Programs Underlying Neuroblastoma Heterogeneity - Aayushi Notra	Improving Phylogenetic Resolution in Large-Scale Microbial Genomic Studies with Soft-Core SNP Filtering - Mona Taouk
14:30	ZMYND8: A novel transcriptional regulator of the dorsal-ventral patterning of the brain - Juli Wang	Lepserico: Working towards genetic typing of leptospira serovars - MJ Sullivan
14:40	Combinatorial multivariate epigenomic patterns reveal the genomic basis of heterogeneity underpinning complex traits and disease - Woo Jun Shim	A Compressive Bayesian Non-negative Matrix Factorization Approach for Detecting Mutational Signatures in Microbes from Phylogenies - Lankamullege Hiruni Navoda Wijewardena
14:50	RNA-seq uncovers consistent sex-driven transcriptional differences in neutrophils across stimuli - Sourav Shyam	ConsensusMetaDA: an R Package for consensus-based differential abundance analysis of metagenomic data - Manoharan Kumar
15:00	Afternoon Tea, Posters & Exhibitor time	
15:50	***William Magarey Room - Spatial & Single Cell.*** *Sponsored by MGI Australia* *Chairs - Nora Liu & Shila Ghazanfar*	***John Halbert Room - Gene Regulation & Epigenetics.*** *Sponsored by Oxford Nanopore* *Chairs - Melanie Smith & Nikeisha Caruana*
	Three Lenses to See Space: Neighborhoods, Density, and Domains in Spatial Transcriptomics - Dr. Ning Liu, SAiGENCI & University of Adelaide	Comprehensive evaluation of approaches for long-read reference-free transcriptome analysis - Feng Yan
		A vision of translational computational pharmacogenomics - Michael Patrick Menden
16:10	Characterising HNSCC and Skin tumour microenvironment using ultra-high plex spatial omics analyses - Chin Wee Tan	PepHLAI: A Novel Machine Learning Tool for Predicting HLA-Binding Peptides - Puthen Veettil Jithesh
16:20	Drug Perturbation Profiling of Altered Cell-Cell Communication Networks in Alzheimer Disease - Seyedeh Sedigheh Abedini	Multimodal framework for ctDNA methylation and mutation modelling in tissue-of-origin prediction - Ruining Dong
16:30	Identifying spatial biomarkers of immunotherapy response in NSCLC using CosMx WTx - Dharmesh D. Bhuva	Genetic regulation of cell type specific chromatin accessibility shapes immune function and disease risk - Angli Xue
16:40	Reconstruction of gene regulatory networks in brain organoids to investigate the role of SUV420H1 during human neurodevelopment - Maria Rosaria Nucera	Multiomics Integration Pipeline for De Novo Epigenetic Analysis in Neurodevelopmental Disorder Trios - Jamie-Lee Thompson
16:50	Capturing cancer fusions at single-cell resolution - Jovana Maksimovic	Characterizing disease in epigenetic machinery: a case for DNA methyltransferases and Histone acetyltransferases - Stephanie Portelli
17:00	Democratizing Single-Cell Foundation Models through Scalable Metacell Aggregation - Muhammad Haroon Shakeel	Estimating rDNA copy number using nanopore sequencing - Zaka Yuen
17:10	Using spatial computational transcriptional tracing to infer cellular origins of the lateral plate mesoderm - Hieu Nim	Adaptively integrated sequencing and assembly of near-complete genomes - Hasindu Gamaarachchi
17:25	**Using historical samples and new methods to study the evolution of plasmids - Prof. Zamin Iqbal, University of Bath.** Sponsored by AMD. Chair - Anna Sheppard.	
18:05	ABACBS 2025 Day 1 close	
18:30	**ABACBS 2025 Conference dinner, Society Awards & 10 year history retrospective.** **Held at the University of Adelaide Courtyard Cloisters**	

Time	Wednesday 26th (Adelaide Oval)	
09:00	**Conference opening**	
09:10	**From Precise Microbiome Genomics to Precision Medicine - Prof. Ami Bhatt, Stanford University.** Sponsored by Adelaide University. Chair - Robert Edwards	
09:40	**Lightning talks.** Chair - Robert Edwards	
10:10	Morning Tea & Poster Session & Exhibitor time	
11:00	***William Magarey Room - Omics and Health.*** *Sponsored by ALIGN* *Chairs - Liza Kretzschmar & Ellis Patrick.*	***John Halbert Room- Methods & Infrastructure.*** *Sponsored by SAGC.* *Chairs - John Salamon & Stevie Pederson*
	Beyond inclusion - Indigenous Empowerment in Genomic Medicine - Prof. Alex Brown, The Kids Research Institute & ANU	*Perfecting Genome Assemblies With Autocycler - Dr. Ryan Wick, University of Melbourne*
11:20	The NCIG Indigenous Variant Library: A Comprehensive Catalogue of Genetic Variation Across Five Australian Indigenous Communities - Kirat Alreja	iplotx visualises any network in Python, with style - Fabio Zanini
11:30	CONCERT-DR: A Context-Aware Transcriptomic Framework for Drug Repositioning - Hengxin Pan	Ecosystem for living benchmark and data-driven method recommendation - Xiaoqi Liang
11:40	Fitting local molecular clocks: a new client-side tool and challenges - Leo Featherstone	ATAC of the Clones: Cancer Clone Identification and Copy Number Estimation from Single-cell ATAC-seq - Lachlan Cain
11:50	Interpreting hTERT Missense Variants in Telomere Biology Disorders Using Machine Learning - Georgina Becerra Parra	LMI4Boltz: Optimizing VRAM utilization to predict the structures of large macromolecular complexes with consumer grade hardware - Thomas Litfin
12:00	A statistical approach for removing joint and modality-specific unwanted variation from single-cell multi-omics data - Hsiao-Chi Liao	SpliceTraceR identifies atypical splicing events driven by splice-site variants using targeted next-generation sequencing data - Muneeza Maqsood
12:10	Linking genomes, environment and disease: A new era for Buruli ulcer surveillance and prevention - Andrew Buultjens	CMDdemux: an efficient single cell demultiplexing method - Jianan Wang
12:20	ALL Aboard the Pan-Genome Express: Enhancing Alignment Precision and Variant Calling in Acute Lymphoblastic Leukaemia Patients - Ashlee Thomson	Building National Infrastructure for Human Genomic Data Sharing and Analytics in Australia - Conrad Leonard
12:30	Six papers, a website and a monograph: Pharmacogenomics for Indigenous Australians - Adam Heterick	MOZART: A Comparative Framework for the Evaluation and Benchmark of Multi-Omics Integration Tools - Guannan Yang
12:40	Lunch & ABACBS Annual General Meeting. AGM will start at approx. 12:55. Chair - Jessica Mar	
	William Magarey Room - AI & Machine Learning. *Sponsored by Intersect* *Chairs - Lan Nguyen & Alex de Sá*	***John Halbert Room - Fundamental Biology.*** *Sponsored by CSL* *Chairs - Emma de Jong & Michael Roach*
13:50	*Transforming RNA-seq into a clinical tool for paediatric oncology - Dr. Chelsea Mayoh, Childrens Cancer Institute, Sydney*	*How bioinformatics helps to constrain global sea level rise projections? - Dr. Sally Lau, James Cook University*
14:00		
14:10	Interpretable deep generative ensemble learning for single-cell omics with Hydra - Manoj M Wagle	Solving Plant Genome Assembly Complexity: Organelles, Haplotypes, and Chromosomes - Chenxi Zhou
14:20	Renal-Agent: A Multimodal Agentic AI for Renal Allograft Biopsy Assessment - Harry Robertson	Rendering Cell Fate in 4D: A Single-Cell Virtual Zebrafish Embryo - Jialin Ding
14:30	treeBayes: Advanced Bayesian Ensemble Learning with Synthetic Data Generation and Comprehensive Uncertainty Estimation for Random Forest - Jinjin Chen	EMUMADZ: Enhanced MUtation Mapping and Detection in Zebrafish - Tyrone Chen
14:40	Developing AI approaches to unravel factors that drive disease relapse in low-risk, early-stage colorectal cancer - Charla Lu Chai	Cell-cycle dependence of bursty gene expression: insights from fitting mechanistic models to single-cell RNA-seq data - Augustinas Sukys
14:50	Beyond the Paradigm: When Foundations Aren't Enough for Spatial and Single-Cell Omics. Benchmarking the Future - Sally Chen	A visual framework for detecting contamination in eukaryotic presence-absence variation pangenomes - Chelsea Matthews
15:00	Using LLM agents to autonomously analyse and integrate public RNA-seq datasets - Kevin Chen	Chromatin-associated TFs shape complex promoter interactions of housekeeping and zinc finger genes - Martin Loza
15:10	Afternoon Tea & Poster Session & Exhibitor Time	
15:40	**Best COMBINE Talk.** Chair - George Bouras	
15:50	**Building the Spatial Tissue Atlas: AI-Powered Integration of Multi-modal Omics Data - A/Prof. Jinmiao Chen, A*STAR.** Sponsored by SAiGENCI. Chair - Mirana Ramialison	
16:20	**Conference awards and Conference close.** Chair - Jimmy Breen	

Tuesday 25th - Lightning Talks (10am)		
Poster Number	Presenter	Title
24	Xixi Li	Decoding the Role of CSF2 (GM-CSF) in Psoriasis via Multi-Omics and Multi-Cohort Analysis
32	Joshua Vescovi	Novel metabolomic signature as a biomarker to detect subclinical atherosclerosis in individuals traditionally classified as low risk.
33	Hettiarachchige Don Jithma Sachini Wijewardena	An Isoform-Aware Hierarchical Random Forest Model for Cell Type Classification in Single-Cell RNA Sequencing.
34	Lei Qin	Estimating cell-cell communication from spatial transcriptomics
37	Xuan Liu	Molecular subgroups in ALS cerebrospinal fluid highlight immune activation and synapto-axonal pathway involvement
39	Lachlan Baer	tadar: an R/Bioconductor package to reduce eQTL noise in differential expression analysis
43	Rebecca Bengtsson	Exploring the Determinants of Spacer Selection in CRISPR-Cas13b
46	Jacob William Ilijev Meyjes-Brown	Using founder effect and in-silico analysis to discover functional metabolic syndrome genes in the Norfolk Island genetic isolate.
52	Renee Green	PholdAPhage - Assembling complete phage structures one protein at a time
124	Yichen Jiang	Reference-Based or Reference-Free? A Flexible Approach to Cell Type Deconvolution in Spatial Transcriptomics
125	Bryan Lye	Integrating Morphological Profiling and AI for Improved Drug Synergy Prediction
126	Sarah Fay Jackson	Pangenome Graphs to Study Structurally Complex and Medically Relevant Loci
127	Monika Mohenska	Spatially charting the developing fat-tailed dunnart

Wednesday 26th - Lightning Talks (9:40am)		
Poster Number	Presenter	Title
3	Michael B. Hall	Genome size estimation from long read overlaps
4	Ahmed, Syed Faraz	Inferring effects of mutations on SARS-CoV-2 transmission from genomic surveillance data
7	Sudaraka Mallawaarachchi	StrainSpy: Strain-level Resolved Inference for Microbiome-Wide Association Studies
9	Denis Bienroth	Automated integration of multi-slice spatial transcriptomics data in 2D and 3D using VR-Omics
11	Bhavika Kumar	Investigating Glioblastoma trajectory from primary to recurrence in a treatment naive surgical cohort
12	Jie Kang	Multi-objective optimisation for biomarker identification, transferability and translation in precision medicine
13	Di Xiao	Refate identifies chemical compounds to target trans-regulatory networks for cellular conversion
16	Kristen Feher	Analysis of STOMICS spatial transcriptomics prostate cancer samples to identify novel metabolic drivers of lethal prostate cancer
17	Yupei You	Benchmarking long-read RNA-sequencing technologies with LongBench: a cross-platform reference dataset profiling cancer cell lines with bulk and single-cell approaches
20	Anne-Lise Gerard	Spatial Multi-Omics Map Out the Tumour-Immune Microenvironment of Paediatric Rhabdoid Tumours
21	Anuradha Wickramarachchi	A platform for continuous data driven clinical decision-making and trusted research environment at scale
28	Emma Rath	Genetic insights into unfavourable cardiovascular outcomes following congenital heart disease surgery
39	PE Melton	KAN Do: Explainable AI for Integrating Genomic, Lipidomic, and Clinical Data in Complex Disease

Tuesday Posters

Poster Number	Presenter	Title
1	Yen-Kai Chen	Disentangling the Gut-Brain Axis through meta-analyses
2	Guan Xue Lee	Characterisation of Novel Resistance Plasmids in Pasteurella multocida from Australian Feedlot Beef Cattle
3	Haiyi Ye	Guiding Computational Design of Antigen Cocktails to Elicit Robust Broadly Neutralizing Antibodies against Hepatitis C Virus
4	Farhan Ameen	Identifying Interface-Enriched Features Between Spatial Domains with FrontiR
5	Chengyi Ma	Benchmarking multivariate statistical methods for integrating imaging and omics datasets
6	Lauren Tjoeka	Characterising resistance with long-reads at single cell resolution in HR deficient ovarian cancer
7	Aaryan Chhabra	Genomic determinants of tissue-specific splicing
8	Oliver Cheng	Nailpolish: Deduplication and error correction for UMI tagged long-read data
9	Tayla Albertini	Single-Cell Modelling of Transcriptional Dynamics Driving High-Risk Mesenchymal-Type Neuroblastoma
10	Peiwen Liu	connectomeDB2025: a rigorously curated, multi-species resource of experimentally supported ligand-ìreceptor interactions
13	Yang Xu	Systematic benchmarking of spatial transcriptomics analysis workflows
15	Daksh Pratap Singh Pamar	GExMix: An Interpolation-Driven Augmentation Strategy for Transcriptome-based Drug Response Modelling in Precision Oncology
20	Bonson Wong	GPU Accelerated Nanopore Basecalling with Open-source Slorado and Openfish
22	Alebachew Messele Kebede	Use of viral proteins as selective tools to delineate the roles of different STAT3 complexes in multifunctional gene expression
24	Xixi Li	Decoding the Role of CSF2 (GM-CSF) in Psoriasis via Multi-Omics and Multi-Cohort Analysis
29	Liyang Fei	Tracking drug resistant clones in ovarian cancer organoids by combining barcoding-based cell tracking with mini bulk RNA-seq
32	Joshua Vescovi	Novel metabolomic signature as a biomarker to detect subclinical atherosclerosis in individuals traditionally classified as low risk.
33	Hettiarachchige Don Jithma Sachini Wijewardena	An Isoform-Aware Hierarchical Random Forest Model for Cell Type Classification in Single-Cell RNA Sequencing.
34	Lei Qin	Estimating cell-cell communication from spatial transcriptomics
36	Jia Wei Tan	LINDTIE: Identify Novel Structural and Splice Variants in Long-read Transcriptome Sequencing
37	Xuan Liu	Molecular subgroups in ALS cerebrospinal fluid highlight immune activation and synapto-axonal pathway involvement

38	Eugenie Lee	Quantifying Uncertainty in Cell Type Annotation with Perturbation-Based Screening
39	Lachlan Baer	tadar: an R/Bioconductor package to reduce eQTL noise in differential expression analysis
42	Reza Ghamsari	Connecting the Transcriptome and Epigenome to Uncover Hidden Insights in AML
43	Rebecca Bengtsson	Exploring the Determinants of Spacer Selection in CRISPR‚ÄìCas13b
46	Jacob William Ilijev Meyjes-Brown	Using founder effect and in-silico analysis to discover functional metabolic syndrome genes in the Norfolk Island genetic isolate.
47	Manoj M Wagle	Deep representation learning of patient-disease heterogeneity in single-cell atlases
48	Inam ul haq azad	Landscape of epitopes targeted by T cells during Hepatitis C virus infection
49	Zunair Khurram	MxSure: Estimating within-host substitution rates and outbreak SNP thresholds with a mixture distribution
50	Alexander Stuart	Novel diversity and ancient persistence of CR1-group retrotransposons across chordates
51	Daniel Kim	Optimising Spatial Transcriptomic Experiments via Multi-criteria Gene Selection and ROI Identification
52	Renee Green	PholdAPhage ‚Äì Assembling complete phage structures one protein at a time
53	Bethany Cross	Promoter Availability Drives Sequence Evolution Upstream of blaKPC Antibiotic Resistance Genes
54	Anusuiya Bora	A deep dive into the reproducibility and reliability of enrichment analysis.
55	Max Martin Docherty-Kenny	A Scalable Framework for Tracking Antimicrobial Resistance Gene Dissemination from a Gene Perspective
56	Harrison A Mueller	Challenges and Opportunities in Spatial Transcriptomics: Insights from Wnt4-KO Kidney Development
57	Oliver John Williams	Flankalyser: A computational pipeline to characterise insertion sequence integration preferences in bacteria
58	Seren Adachi	Identifying and validating protein biomarker signatures from tear-fluid of Amyotrophic Lateral Sclerosis patients
59	Nicolo Fabila	Investigating the Therapeutic Potential of Psilocybin for Compulsive Eating in a Rat Model of Binge Eating
60	Zachary Riedlshah	Learning Latent Chromatin Regulatory States Across Genomic Scales
61	Jingjing Liu	Mapping Dengue T Cell Epitope Landscape to Inform Targets for Vaccine Design
62	Amrita Acharya	Molecular modeling studies of phytochemicals from cannabis sativa acting on cannabinoid CB2 receptor as a potential anti-Alzheimer agent
63	Jia Yi Hee	Peripheral blood gene expression classifies progression to type 1 diabetes in autoantibody-positive children

64	Rojashree Jayakumar	Quantifying Subcellular Compartment-Specific Cell‚ÄìCell Communication Using a Hierarchical Machine Learning Framework
65	Mingyan Fang	Stereopy: modeling comparative and spatiotemporal cellular heterogeneity via multi-sample spatial transcriptomics
66	Linpeng Xie	Structure-Guided Computational Drug Repurposing for Duchenne Muscular Dystrophy
67	Christiaan Johannes van Aardt	Assessing Variability and Robustness of Spatial Transcriptomic Features Across Diverse Datasets
69	Ching Yin Wan	IsoVis: a website to visualize and analyze alternative RNA isoforms
70	Calandra Vanessa Grima	Profiling technical biases and custom probe performance in 10X Flex scRNA-seq for single-cell detection of cancer fusion transcripts
71	Ishika Mahajan	RNA-Seq Driven Immunogenomic Profiling of Pediatric Cancer Models Reveals Subtype-Specific Antigen Presentation and Therapeutic Vulnerabilities
72	Kerry John Heffernan	Segmenting Organoids: Evaluating Classical, Specialised, and Generalist AI Approaches for Brightfield Microscopy
73	Qiyang Wu	Computational Integration of Human Gut Metabolomics and Immune Screening for Macrophage Polarisation Analysis
74	Sadia Hasan	Dissecting immune escape mechanisms in colorectal cancer orthotopic organoid mouse models using next-generation sequencing
75	Suet Li Hooi	Post-antibiotics Gut Microbiome Dynamics: Temporal computational modelling of Microbiome Restoration and Resilience
76	Amelia Dunstone	Software tools for the analysis of multi-state and multi-omic QTL data
77	Sasuni D. Hirimuthugoda	Using Graph Theory to Identify Functional Units in the Spatial Distribution of Cells in Cancer Tissues
78	Munazzah Maqbool	What's driving the spread of third-generation cephalosporin resistance in clinical E. coli?
79	Elham Shokati	A Single Cell Atlas of Cytokine-induced killer cells derived from colorectal cancer patients
80	Hao Huang	Identifying causal variants underlying GWAS and single-cell multi-layer QTLs with multi-phenotype fine-mapping
81	Kristy Nguyen	In-depth immunological risk assessments and predictive modelling to optimise kidney transplant outcomes
82	Sichang Ding	Multi-task benchmarking of single-cell multimodal omics integration methods
83	Vladimir Shikov	SVEnsemble2: a Machine Learning-based Integrative Framework for Detection of Structural Variation from High Throughput Sequencing Data
84	Gemma Laird	Unveiling the Lifestyle of Phylum Bipolaricaulota Members in Global Ecosystems

85	Yulin Wu	A pipeline for detecting differential RNA modification at the isoform resolution using direct RNA sequencing data
86	Haiyi Ye	AlphaFold3-Enabled Discovery of Early Infection Antibody Targets in an HCV Super-Clearer
87	Roopan Giri	Integrative Analysis of microRNA, circular RNA, and Alternative Splicing in Neuroblastoma
88	Yitong Yao	Health Economic Modelling of Post Kidney Transplant Screening Biomarkers
90	Max Woollard	Weighted sliced inverse regression for scalable supervised dimensionality reduction of spatial transcriptomics data
91	Nikita Dalvi	Quantifying treatment response in Mucinous Ovarian Cancer Organoids using 3D image-based analysis and machine learning methods.
92	Coco Xiaoge Huang	A Deep Transfer Learning Model for Predicting Uterine Carcinosarcoma Recurrence with Multi-Modal Data Integration
93	Fatemeh Shamsesfandabadi	AI-Driven Prediction of Clinical Trial Success for Antibody Therapeutics.
94	Caitlin Gwendoline Page	Investigating the methylation and fragmentomic properties of cell-free DNA using Nanopore sequencing
95	Minghan Wu	Revealing the landscape and characteristics of trans-splicing in health and cancer
96	Taha Ozair Osman	The Role of Timing in Combination Therapy: How Network Motifs Shape Drug Response
98	Tricia S.E Lee	Long reads improve tracking of OXA181 resistant plasmid over 7 years
99	Xiang Guo	Understanding how networks of cell communication in cancers lead to phenotypic diversity
100	Zakir Hossine	Network-Based Differential Abundance Analysis Enhanced by Thresholded Gaussian Copula Simulation.
101	Van Nguyen Hoang	Impact of phylogenetic models using RNA secondary structures: A survey on Rfam database.
102	Galen Raphael Pereira	Interpretable Artificial Intelligence Models for Cellular Insights and Clinical Impact
103	Zeynep Jennifer March	Accurate detection of exon-exon boundaries in noisy Oxford Nanopore direct RNA sequencing data
104	Michelle Meier	Exploring biologically meaningful doublets in single-cell transcriptomics
105	Luis A. Arriola	Identifying Germline Structural Variants in Paediatric Acute Myeloid Leukaemia
106	Puning Wang	Non-Stationary Evolution as a Driver of Molecular Clock Violation
107	Chiara Folland	Realignment of genomic short-read sequencing data from unsolved rare disease patients to T2T-CHM13 and the draft human pangenome
108	Yifan Wang	riboHMM2: a hidden Markov model for the identification of translational regulation elements in mRNAs from ribosome profiling data

109	Huize Gao	Tracing ancient viral epidemics using virus-interacting proteins
110	Deannon Branch	Using Metagenomics to Investigate Antimicrobial Resistance Plasmids in Biosolids-Amended Soil
111	Jennifer Ureta	Analytical methods for ecDNAs in Cancer
112	Namuhan	Detecting milk fat globules in human milk single-cell RNA-sequencing data: a novel computational method
113	Chen Gong	Multi-Omics analysis and AI-Driven Digital Twins for Precision Medicine of Diabetes Kidney Disease
114	Muhammad Jahanzeb	Uncovering the Relationship of Farm Environment and Antimicrobial Resistance
115	James Zuiani	Characterising The Cellular Landscape And Signalling Pathways In Hereditary Pancreatitis Using Spatial Transcriptomics
116	Wenjie Shan	Advanced Bioinformatics for Cross-Biofluid Metabolomics in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome
117	Hashara Kumarasinghe	IQ-TREE GPU: Accelerating phylogenetic inference using GPUs
118	Joshua Mitchell	Improving the fidelity and scope of microbial ecology using bioinformatics, machine learning, and global data
120	Mariana Velasque	GEMOmics: A Fully Open-Source Workflow for Omics-Driven, Condition-Specific Metabolic
123	Tuomas Borman	Orchestrating Microbiome Analysis with Bioconductor
124	Yichen Jiang	Reference-Based or Reference-Free? A Flexible Approach to Cell Type Deconvolution in Spatial Transcriptomics
125	Bryan Lye	Integrating Morphological Profiling and AI for Improved Drug Synergy Prediction
126	Sarah Fay Jackson	Pangenome Graphs to Study Structurally Complex and Medically Relevant Loci
127	Monika Mohenska	Spatially charting the developing fat-tailed dunnart
128	Bridie Moy	Understanding the functional consequences of chronic kidney disease-associated structural variants in Indigenous Australians
129	Khoa Nguyen	Harnessing Machine Learning to Design High-Efficacy CRISPR-Cas13b Guides
130	Raymond Louie	Inferring Drivers of Viral Evolution from Longitudinal Sequencing Data Using a Mathematical Framework

Wednesday Posters

Poster Number	Presenter	Title
1	Xiaochen Zhang	CellDiffusion: a generative model to annotate spatial and single cell RNA-seq from bulk references
2	Leo Featherstone	How does date-rounding affect phylodynamic inference for public health?
3	Michael B. Hall	Genome size estimation from long read overlaps
4	Ahmed, Syed Faraz	Inferring effects of mutations on SARS-CoV-2 transmission from genomic surveillance data
5	Ahmed, Syed Faraz	Comparative Genomic Assessment of Vaccine Effectiveness Across Emerging Mpox Clades
6	Wytamma Wirth	Snippy-NG: The Next Generation of Microbial Variant Calling
7	Sudaraka Mallawaarachchi	StrainSpy: Strain-level Resolved Inference for Microbiome-Wide Association Studies
8	Wytamma Wirth	The Centre for Pathogen Genomics Bioinformatics Analysis Portal
9	Denis Bienroth	Automated integration of multi-slice spatial transcriptomics data in 2D and 3D using VR-Omics
10	Sungyoung Shin	Dynamic mechanistic modelling predicts and validates synergistic and sequential drug strategies to overcome cancer resistance
11	Bhavika Kumar	Investigating Glioblastoma trajectory from primary to recurrence in a treatment naive surgical cohort
12	Jie Kang	Multi-objective optimisation for biomarker identification, transferability and translation in precision medicine
13	Di Xiao	Refate identifies chemical compounds to target trans-regulatory networks for cellular conversion
14	William James Ferraro Rieger	Squidly: Enzyme Catalytic Residue Prediction Harnessing a Biology-Informed Contrastive Learning Framework
15	Claire Cheng	A CLIP-based foundation model aligns mouse and human brain single-cell transcriptomes for cross-species discovery
16	Kristen Feher	Analysis of STOMICS spatial transcriptomics prostate cancer samples to identify novel metabolic drivers of lethal prostate cancer
17	Yupei You	Benchmarking long-read RNA-sequencing technologies with LongBench: a cross-platform reference dataset profiling cancer cell lines with bulk and single-cell approaches
19	Ahmed Abdul Quadeer	Seasonal antigenic prediction of influenza A H3N2 using machine learning
20	Anne-Lise Gerard	Spatial Multi-Omics Map Out the Tumour-Immune Microenvironment of Paediatric Rhabdoid Tumours
21	Anuradha Wickramarachchi	A platform for continuous data driven clinical decision-making and trusted research environment at scale
22	Anuradha Wickramarachchi	Accelerating Analysis on UK Biobank 500,000 WGS: RAPpoet Boosts Speed and Cut Costs
23	Diya Prabhuram	AI-Driven Integrative Frameworks to Advance Precision Therapeutics for Retinopathies

24	Halimat Chisom Atanda	Associating cancer-specific single nucleotide variants with methylation changes using convolutional neural networks
25	Lahiru Sandaruwan Boyagoda Wasala Arachchige	Does Field-of-View Affect Imaging-Based Spatial Transcriptomics?
28	Emma Rath	Genetic insights into unfavourable cardiovascular outcomes following congenital heart disease surgery
31	Sagrika Chugh	What happens to the bone marrow niche after cancer therapy? We asked 114,000 cells.
32	Sebastian Hollizeck	Adaptive Sampling Nanopore Sequencing: Promise and Pitfalls in Clinical Targeting
33	Chau Nhi Nguyen	Benchmarking Structural Variant Imputation Accuracy Across Diverse Human Ancestries Using Short-Read Data
34	Andrew Lonsdale	Browser-Based Bioinformatics Breaks Barriers, Boosts Brilliance, Brings Big Benefits
35	Chen Zhan	Deliver Human Cell Atlas via pre-trained models using Bayesian Inference
36	Tyrone Chen	EMUMADZ: Enhanced MUtation Mapping and Detection in Zebrafish
37	Urwah Nawaz	Enhancing Neuroblastoma Risk Stratification Through Proteomic Signatures
39	PE Melton	KAN Do: Explainable AI for Integrating Genomic, Lipidomic, and Clinical Data in Complex Disease
40	Alexandre Xavier	Know what to expect: An atlas of co-occurring ICD10 disease codes
41	Andreas Bachler	Machine-learning genome annotation tools do a surprisingly good job in some non-model organisms and can even help spot mis-annotated genes!
42	Shreya Rajesh Rao	Making Space Count: Strategies for Analysing Spatial Omics Data
43	Malathi Imiyage Dona	Sex-specific drivers of cardiac remodelling in disease
44	Joel Jeffrey Duncan Moffet	SMINT: An integrative spatial multi-omic workflow for unified analysis of tumour tissue
45	Jocelyn Hume	The spread of IMP-4 globally and in Australia
46	David Chisanga	A multi-layered transcriptional circuitry regulates the differentiation of effector regulatory T cells
47	Phoebe Carr	Comparison of high-throughput whole genome sequencing methods for bacteriophages
48	Kerry A. Mullan	CRISPR-Cas13b Guide Discovery Platform for Targeting RNA Virus Families
49	Gunjan Dixit	Growing Fast, changing Faster: Rapid developmental changes in a multi-tissue pediatric airway atlas
50	Shian Su	Investigating X-inactivation escape in immune cells of female lupus patients using long-read single-cell sequencing

51	Yue Cao	Large language models for interdisciplinary communication in bioinformatics
52	Roxana Zahedi	OmniST: Toward a Foundation Model-Based Graph Transformer in Spatial Transcriptomics
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54	Aaron Wing Cheung Kwok	Removing contamination from Image-based Spatial Transcriptomics data with DenoIST
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56	Luan Zhong	Unmasking the jumpers: Machine learning for fast and automated transposons annotation
57	Xinyi Jin	Where are the batch effects in my spatial data? Comprehensive spatial profiling of human kidney organoids and foetal kidney
58	Magdalena Antczak	Beyond the pipeline: building bioinformatics workflows for non-bioinformaticians
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60	Geraldine Kong	From MAGs to Mechanisms: Community Metabolic Modelling Predicts Functional Consequences of COVID-19 Gut Microbiome Changes
61	Changqing Wang	SpatialBench: Comparative benchmarking of high-resolution spatial transcriptomics platforms using reference tissues
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63	Pratibha Panwar	A systematic technical review of 45 spatial omics clustering methods reveals trends and gaps in analysis techniques.
64	Bree Tillett	Changes in gut inflammation and permeability are reflected in host gut barrier proteins
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66	Beata Kiedik	Exploring cellular heterogeneity of localised luminal breast cancers
68	Abdul Munif	Pairwise Ranking based on Histone Modifications
69	Laura K Inglis	The potential for machine learning to classify metagenomes by isolation source
71	Surafel Tegegne	Exercise EV Proteomics Explorer: A Hybrid Web Platform for Curated Data Mining and Live Functional Analysis of Extracellular Vesicle Proteins
72	Mansi Aggarwal	Exploring microRNA-mRNA Relationships Mediated by Transcription Factors in Cancer
74	Nozhat Hassan	Lineage-Specific Evolution, Structural Diversity, and Activity of R2 Retrotransposons in Animals
79	Fabio Zanini	Using AI to hold cells, living and extinct
80	Roshan Jalaldeen	Multiplex immunohistochemistry spatial proteomics analysis workflow: Fluorescence ANalysis for Tissue Cytometry (FANTCY)
81	Alexandra Gavryushkina	Evaluating methods for reconstructing cancer cell lineage trees from single-cell RNA-seq data

82	Emma de Jong	Determining the optimal strategy for homology-based genome annotation in bony fish (Class Actinopterygii)
83	Adriana Zanca	Electric feel: a computational investigation of electric moments and shielding in biological cells
84	Jianan Fan	Cell Type-Specific Chromatin Accessibility Links Genetic Regulation to Immune Function and Disease Risk Beyond eQTLs
85	David Ma	Hybrid HPC‚ÄìCloud Storage for Bioinformatics: Fast Local Analysis, Scalable Archival, and Open Infrastructure-as-Code Tools from the Australian Genome Research Facility
86	Mei Du	A multi-analytical approach using single-cell sequencing to identify early life immune pathways defining peanut allergy outcomes in childhood
87	Dania Al Istwani	Adapting scRNA-seq workflow to explore Ntrk2's transcript isoforms
88	Nikeisha Caruana	Advancing Inclusion, Diversity, and Accessibility in Australian Bioinformatics
89	Sabrina Silveira	BENDER DB: a database of protein binding sites across neglected disease proteomes
90	Anna Balzer	Deconvolution of bulk RNAseq from cerebellum and hippocampus in a pre-clinical model of repetitive mild-traumatic brain injury
91	Steven Morgan	Evaluating Cross-Species Cell Type Annotation in Single-cell RNA-sequencing Data
92	Kara Henry-Cocks	Human genetic library sequence verification using short-read sequencing and de novo assembly
93	Cong Pham	Integration of the BacDive database into metagenomic analyses with taxon set enrichment analysis
94	Anna Li	Microbiome chimerism: A new frontier in GvHD pathogenesis?
95	Stephen Pederson	motifTestR: A Bioconductor package for testing motif over-representation or positional bias
96	Linh Hang To	Multi-Omics Prediction of Chemotherapy Side Effects: Early Insights from the PREDiCT Biobank
97	Luke Garratt	Near complete genome sequences of the pathogenic fungi Scedosporium aurantiacum and Scedosporium apiospermum
98	Alistair Chalk	Parallel proteomics and transcriptomics reveal post-transcriptional remodelling of in vivoCD8 T cell immunity to acute and chronic viral infection
99	Caitlin A. Selway	PHEvir: an automated nextflow pipeline to simultaneously characterise multiple viral pathogens from ONT amplicon data
100	Alex G. C. de Sa	Robust and Explainable Automated Machine Learning (AutoML) for Biochemical Property Prediction
101	Lucia Rey Alvarez	Sex-specific patterns of nuclear insertion of mitochondrial DNA
102	Imalki Kariyawasam	Spatial Dynamics of the Tumour Immune Micro-environment in HGSOC: Insights from the SOLACE2 Trial
103	Stefanie Navaratnam	Thanks CATHY! Semi-automating the Annotation of High Dimensional Cytometry Data
104	Alexandra Sneddon	Utilising biological knowledge as a prior for resource-efficient LLM training

105	Marek Cmero	SODA-Hub provides an end-to-end analysis environment for spatial omics from data ingestion to cell segmentation
106	Vijini Mallawaarachchi	agtools: a software framework to manipulate assembly graphs
107	Mengyuan Shen	cellNexus: Multi-resolution curated representation of the Human Cell Atlas to accelerate discovery
108	Ignatius Jeppe Menzies	CTRL: dynamic consent platform to support national-scale genomics research
109	Mandeep SIngh	Computational Analyses of Nature Variants from Arabidopsis Thaliana Paired NLRS (RPS4:RRS1) Linked to Thermotolerance
110	Dr. Katari Sudheer Kumar	Integrative Computational Pipeline for Targeting the LTK L650F Mutation: Multi-Scale Screening, ADMET, Quantum, and Molecular Dynamics Approaches
111	Juan Salvador Inda Diaz	How Antibiotic Resistance Gene Detection Tools Shape Our View of the Resistome
112	Raman Sethi	ezChat: An Automated Chatbot based Single-cell and Spatial Omics Analysis Solution for Bench Scientists
113	Jake Bradford	Approximating Genome-Wide CRISPR Off-Target Risk with a Compact Index
114	Ove Johan Ragnar Gustafsson	WorkflowHub: share, describe, and credit computational workflows
115	Mahdieh Mahdavi	Identifying gene fusions in single cell RNA sequencing
116	John Reeves	qxub: Frictionless, reproducible HPC job submission for bioinformatics
117	James Douglas	BioSpec: Research project specifications to guide agent-assisted computational biology
118	Jenny Draper	BactoPipe: a YAML-configured, workflow-lite framework for flexible, auditable bacterial genomics on small servers
119	Jenny Draper	Genomic diversity of carbapenem-resistant Serratia marcescens in an ICU
120	George Bouras	Baktfold - Ultra-sensitive Protein Structure-Informed Functional Annotation of Hypothetical Proteins in Bacteria
121	Hiruna Samarakoon	Leveraging basecaller's move table to generate a lightweight k-mer model for nanopore sequencing analysis
122	Inam ul haq azad	IFNBoost: An interpretable computational model for identifying IFN inducing peptides